Common components of OP: decision; objective; constraints. Feasible solution: a decision that satisfies all the constraints. Feasible region: set of feasible solutions. Optimal solution: the (feasible) decision variable that attains as least as good objective values as other feasible solutions.

Shortest path problem: E: the set of edges, Wij: the distance between node i and node j.

For each edge $(i,j) \in E$, define

$$x_{ij} = \begin{cases} 1 & \text{if we use edge } (i,j) \\ 0 & \text{otherwise} \end{cases}$$

An optimization model:

$$\begin{array}{ll} \text{minimize} & \sum_{(i,j)\in E} w_{ij}x_{ij} \\ \text{subject to} & \sum_j x_{sj} = 1 \\ & \sum_j x_{jt} = 1 \\ & \sum_j x_{ij} = \sum_j x_{ji}, \quad \forall i \neq s,t \\ & x_{ij} \in \{0,1\}, \quad \forall (i,j) \in E \end{array}$$

Vertex cover problem: a graph consists of nodes V and edges E, find the smallest set of vertices that touch every edge of the

Define the following decision variables:

$$x_i = \begin{cases} 1 & \text{if we choose vertex } i \\ 0 & \text{otherwise} \end{cases}$$

An optimization model can be written as:

$$\begin{array}{ll} \text{minimize} & \sum_i x_i \\ \text{subject to} & x_i + x_j \geq 1, \quad \forall (i,j) \in E \\ & x_i \in \{0,1\} \quad \forall i \in V \end{array}$$

Support vector problem: find a separation plane $\mathbf{a}^T\mathbf{x} + b = 0$.

Define $\delta_i = (1 - \mathbf{a}'\mathbf{x}_i - b)^+$ and $\sigma_j = (\mathbf{a}'\mathbf{y}_j + b + 1)^+$.

We can first write this as

$$\begin{array}{ll} \text{minimize}_{\mathbf{a},b} & \sum_i \delta_i + \sum_j \sigma_j \\ \text{subject to} & \delta_i = (1 - \mathbf{a}^T\mathbf{x}_i - b)^+, \quad \forall i \\ & \sigma_j = (\mathbf{a}^T\mathbf{y}_j + b + 1)^+, \quad \forall j \end{array}$$

We claim we can relax "=" to "≥" (why?):

$$\begin{array}{ll} \text{minimize}_{\mathbf{a},b} & \sum_i \delta_i + \sum_j \sigma_j \\ \text{subject to} & \delta_i \geq (1 - \mathbf{a}^T\mathbf{x}_i - b)^+, \quad \forall i \\ & \sigma_j \geq (\mathbf{a}^T\mathbf{y}_j + b + 1)^+, \quad \forall j \end{array}$$

equivalent to sigma > = both of 0 and 1-aTxi -b, which transforms it to linear

$$\begin{array}{ll} \text{minimize}_{\mathbf{x}} & \mathbf{c}^T\mathbf{x} \\ \text{subject to} & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq 0 \end{array}$$

Eliminating inequality constraints $A\mathbf{x} \leq \mathbf{b}$ or $A\mathbf{x} \geq \mathbf{b}$

- Write it as $A\mathbf{x} + \mathbf{s} = \mathbf{b}, \mathbf{s} \geq 0$, or $A\mathbf{x} - \mathbf{s} = \mathbf{b}, \mathbf{s} \geq 0$
- We call $\mathbf{s}$ the slack variables

If one has $x_i \leq 0$

- Define $y_i = -x_i$

Eliminating "free" variables x_i (no constraint on x_i)

- Define $x_i = x_i^+ - x_i^-$, with $x_i^+ \geq 0$, $x_i^- \geq 0$
- Define $y = \max_{i=1,\ldots,n}\{\mathbf{c}_i^T\mathbf{x} + d_i\}$

$$\begin{array}{ll} \text{minimize}_{\mathbf{x},y} & y \\ \text{subject to} & y \geq \mathbf{c}_i^T\mathbf{x} + d_i \quad \forall i \end{array}$$

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^n |x_i| \\ \text{s.t.} & A\mathbf{x} = \mathbf{b} \end{array}$$

lently written as

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^n y_i \\ \text{s.t.} & y_i \geq x_i \\ & y_i \geq -x_i \\ & A\mathbf{x} = \mathbf{b} \end{array}$$

Definition

A *polyhedron* is a set that can be described in the form

$$\{\mathbf{x} \in \mathbb{R}^n | A\mathbf{x} \geq \mathbf{b}\}$$

where A is an $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$.

A set $S \subseteq \mathbb{R}^n$ is *convex* if for any $\mathbf{x}, \mathbf{y} \in S$, and any $\lambda \in [0,1]$, $\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in S$.

For any $\mathbf{x}_1, \ldots, \mathbf{x}_n$ and $\lambda_1, \ldots, \lambda_n \geq 0$ satisfying $\lambda_1 + \cdots + \lambda_n = 1$, we call $\sum_{i=1}^n \lambda_i\mathbf{x}_i$ a convex combination of $\mathbf{x}_1, \ldots, \mathbf{x}_n$.

Let P be a polyhedron. A point $\mathbf{x} \in P$ is said to be an extreme point of P if we cannot find two vectors $\mathbf{y}, \mathbf{z} \in P$, both different from $\mathbf{x}$, and a scalar $\lambda \in [0,1]$, such that $\mathbf{x} = \lambda\mathbf{y} + (1-\lambda)\mathbf{z}$

That is, $\mathbf{x}$ cannot be represented as a convex combination of other points in P.

Theorem: for the standard LP polyhedron {x: Ax =b, x>=0} the followings are equivalent: 1. x is an extreme point; 2. x is a basic feasible solution.

Theorem (LP fundamental theorem): given a linear optimization problem where A has full row rank m. (1) If there is a feasible solution, there is a basic feasible solution. (2) If there is an optimal solution, there is an optimal solution that is a basic feasible solution.

Corollary: (1) in order to find an optimal solution, only need to search among basic feasible solutions. (2) if an LP with m constraints (in the standard form) has an optimal solution, then there must be an optimal solution such that there are no more than m positive entries.

Two basic solutions are neighboring if they differ exactly one basic (or non-basic) index. In a basic solution, there are no more than m positive entries. There are finitely many basic solutions for a given LP. if a LP gas a unique optimal solution, then it must be a BFS.

Theorem (Stopping Criterion): consider a basic feasible solution x associated with the basis B(1), ...,B(m) and let c_bar be the corresponding vector of reduced costs. If c_bar >=0, then x must be optimal.

We start from a BFS $\mathbf{x}$ (with corresponding basis B)

1. We first compute the *reduced costs* $\bar{\mathbf{c}}$

$$\bar{c}_j = c_j - \mathbf{c}_B^T A_B^{-1} A_j$$

   - If no reduced costs is negative, then $\mathbf{x}$ is already optimal
   - Otherwise choose some j such that $\bar{c}_j < 0$
2. Compute the jth basic direction $\mathbf{d} = [-A_B^{-1}A_j; 0; \ldots; 1; \ldots; 0]$
   - If $\mathbf{d} \geq 0$, then the problem is unbounded, i.e., the optimal value is $-\infty$.
   - Otherwise, compute $\theta^* = \min_{i \in B, d_i < 0}\{-\frac{x_i}{d_i}\}$
3. Let $\mathbf{y} = \mathbf{x} + \theta^*\mathbf{d}$. Then $\mathbf{y}$ is the new BFS with index j replacing $B(\ell)$ in the basis, where $B(\ell)$ is the index attaining the minimum in θ^*. Objective value is changed by $\theta^*\mathbf{c}^T\mathbf{d} = \theta^*\bar{c}_j$.
4. Repeat these procedures

Degeneracy: for a basic feasible solution, some of the basic variables are 0, equivalently to the number of non-zeros at a basic solution is strictly less than m.

Theorem (Bland's Rule): if we used both the smallest index rule for choosing the entering basis and the exiting basis, then no cycle will occur in the simplex algorithm.

In the two-phase simplex method, we first solve an auxiliary problem (**e** means an all-one vector).

$$\begin{array}{ll} \text{minimize}_{\mathbf{x},\mathbf{y}} & \mathbf{e}^T\mathbf{y} \\ \text{subject to} & A\mathbf{x} + \mathbf{y} = \mathbf{b} \\ & \mathbf{x}, \mathbf{y} \geq 0 \end{array}$$

Without loss of generality, we assume $\mathbf{b} \geq 0$ (otherwise, we pre-multiply that row by -1).

There is a trivial BFS to the auxiliary problem: $(\mathbf{x} = 0, \mathbf{y} = \mathbf{b} \geq 0)$ so one can apply the Simplex method to solve it.

Theorem

The original problem is feasible if and only if the optimal value of the auxiliary problem is 0.

Phase I:

1. Construct the auxiliary problem such that $\mathbf{b} \geq 0$
2. Solve the auxiliary problem using the Simplex method
   - If we reach an optimal solution with optimal value greater than 0, then the original problem is infeasible
3. If the optimal value is 0 with optimal solution $\mathbf{x}^*$, then we enter phase II

Phase II: Solve the original problem starting from the BFS $\mathbf{x}^*$

- If $\mathbf{x}^*$ is degenerate, then we need to supplement some indices to make it a BFS for the original problem

$\mathbf{c}^T - \mathbf{c}_B^T A_B^{-1} A$	$-\mathbf{c}_B^T A_B^{-1}\mathbf{b}$
$A_B^{-1}A$	$A_B^{-1}\mathbf{b}$

The lower part of the tableau can be viewed as a transformation of the constraint $A\mathbf{x} = \mathbf{b}$ to $A_B^{-1}A\mathbf{x} = A_B^{-1}\mathbf{b}$

- It is equivalent to the original constraint

Furthermore, if we write $A = [A_B, A_N]$, then

$$A_B^{-1}A = [I, A_B^{-1}A_N]$$

Therefore, this part must contain an identity matrix.

Also when the basis is B, the current basic feasible solution is

$$\mathbf{x} = [\mathbf{x}_B; \mathbf{x}_N] = [A_B^{-1}\mathbf{b}; 0]$$

Therefore the lower right corner gives the current BFS.

$$\mathbf{c}^T - \mathbf{c}_B^T A_B^{-1} A$$

is the reduced cost at this basis.

$$-\mathbf{c}_B^T A_B^{-1}\mathbf{b} = -\mathbf{c}_B^T\mathbf{x}_B$$

is the negative of the objective value at this basis.

$\mathbf{0}_m$	$\mathbf{c}_N^T - \mathbf{c}_B^T A_B^{-1} A_N$	$-\mathbf{c}_B^T\mathbf{x}_B$
$\mathbf{I}_m$	$A_B^{-1}A_N$	$\mathbf{x}_B$

Here $\mathbf{0}_m$ is a vector of m zeros and $\mathbf{I}_m$ is the m-dimensional identity matrix.

This form of LP is called the *canonical* form.

- The constraint matrix for the basic variables (not necessarily the first m columns) is an identity matrix
- The objective part for the basic variables is zeros

Therefore, in the simplex tableau, this is to find

$$\theta^* = \min\left\{\frac{\bar{b}_i}{\bar{A}_{ij}} : \bar{A}_{ij} > 0\right\}$$

where $\bar{\mathbf{b}}$ is the lower right column and $\bar{A}$ is the lower left part of the tableau.

Then we perform the following two steps

1. Divide each element in the pivot row by the pivot element
2. Add proper multiples (could be negative) of the pivot row (after the first step) to each other rows, including the top row of objective coefficients, such that all other elements in the pivot column become zeros (including the top row)
3. Both operations include the right-hand-side column of $\mathbf{b}$

Phase I:

- The auxiliary problem is not in a canonical form: Although the constraint contains an identity matrix (which is the initial basis), the corresponding objective coefficients are not 0's
- We need to compute the top row (reduced costs)
  - For basic part, the reduced costs are 0
  - For nonbasic part, $\bar{c}_j = c_j - \mathbf{c}_B^T A_B^{-1} A_j = -\mathbf{e}^T A_j$, so the jth reduced cost is the negative of the sum of that column
- This also applies to the initial objective value, which equals the negative of the sum of the right hand side vector.

Then to carry out Phase II in the Simplex tableau:

1. Drop all the columns associated with the auxiliary variables
2. Change the top row to the reduced costs corresponding to the original problem by using the reduced cost formula:

$$\bar{c}_j = c_j - \mathbf{c}_B^T A_B^{-1} A_j$$

   Also compute the current objective value
3. Continue the simplex tableau to solve the original problem

- $\mathbf{a}_i^T$ is the ith row of A, A_j is the jth column of A
- Each primal constraint corresponds to a dual variable
- Each primal variable corresponds to a dual constraint
- Equality constraints always correspond to free variables

Primal	minimize	maximize	Dual
constraints	$\geq b_i$	≥ 0	variables
	$\leq b_i$	≤ 0	
	$= b_i$	free	
variables	≥ 0	$\leq c_j$	constraints
	≤ 0	$\geq c_j$	
	free	$= c_j$	

Adding a variable (the rest are kept the same):

- The original BFS is still a BFS, the reduced cost is unchanged
- Only need to check the reduced cost corresponding to the new variable. If it is non-negative, then the original optimal solution is still optimal; otherwise continue the simplex method from there

Adding a constraint:

- If the original optimal solution satisfies the constraint, then it is still optimal
- If not, then the best way to deal with it is to think it as adding a dual variable, then use the simplex tableau for the dual problem to continue calculations

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Primal			Dual		
minimize	$\mathbf{c}^T\mathbf{x}$		maximize	$\mathbf{b}^T\mathbf{y}$	
subject to	$\mathbf{a}_i^T\mathbf{x} \geq b_i$,	$i \in M_1$,	subject to	$y_i \geq 0$,	$i \in M_1$
	$\mathbf{a}_i^T\mathbf{x} \leq b_i$,	$i \in M_2$,		$y_i \leq 0$,	$i \in M_2$
	$\mathbf{a}_i^T\mathbf{x} = b_i$,	$i \in M_3$,		y_i free,	$i \in M_3$
	$x_j \geq 0$,	$j \in N_1$,		$A_j^T\mathbf{y} \leq c_j$,	$j \in N_1$
	$x_j \leq 0$,	$j \in N_2$,		$A_j^T\mathbf{y} \geq c_j$,	$j \in N_2$
	x_j free,	$j \in N_3$,		$A_j^T\mathbf{y} = c_j$,	$j \in N_3$

$$\begin{aligned}\text{minimize}_{\mathbf{a},b,\boldsymbol{\delta},\boldsymbol{\sigma}} \quad & \textstyle\sum_{i=1}^n \delta_i + \sum_{j=1}^m \sigma_j \\ \text{subject to} \quad & \mathbf{x}_i^T\mathbf{a} + b + \delta_i \geq 1, \quad \forall i = 1,\dots,n \\ & \mathbf{y}_j^T\mathbf{a} + b - \sigma_j \leq -1, \quad \forall j = 1,\dots,m \\ & \delta_i \geq 0,\ \sigma_j \geq 0, \quad \forall i = 1,\dots,n,\ \ j = 1,\dots,m\end{aligned}$$

Associate w_i to each of the first set of constraints, v_j to each of the second set of constraints. (Suppose $\mathbf{a}$ is a d-dimensional vector.)

$$\begin{aligned}\text{maximize}_{\mathbf{w},\mathbf{v}} \quad & \textstyle\sum_{i=1}^n w_i - \sum_{j=1}^m v_j \\ \text{subject to} \quad & \textstyle\sum_{i=1}^n x_{ik} w_i + \sum_{j=1}^m y_{jk} v_j = 0, \quad \forall k = 1,\dots,d \\ & \textstyle\sum_{i=1}^n w_i + \sum_{j=1}^m v_j = 0 \\ & 0 \leq w_i \leq 1, \quad \forall i = 1,\dots,n \\ & -1 \leq v_j \leq 0, \quad \forall j = 1,\dots,m\end{aligned}$$

we let $X = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]$ and $Y = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_m]$, then the first constraint can also be written as

$$X\mathbf{w} + Y\mathbf{v} = \mathbf{0}$$

Theorem

If we transform a linear program to an equivalent one (such as by replacing free variables, adding slack variables, etc), then the dual of the two problems will be equivalent.

Theorem

If we transform the primal to its dual, then transform the dual to its dual, then we will obtain a problem equivalent to the primal problem, that is, the dual of dual is the primal.

Theorem (Weak Duality Theorem)

If $\mathbf{x}$ is feasible to the primal and $\mathbf{y}$ is feasible to the dual, then

$$\mathbf{b}^T\mathbf{y} \leq \mathbf{c}^T\mathbf{x}$$

If the primal is a minimization and dual is a maximization, then

- Any dual feasible solution will give a lower bound on the primal optimal value
- Any primal feasible solution will give an upper bound on the dual optimal value
- The optimal value of primal is larger than that of dual
- *If the primal problem is unbounded (i.e., the optimal value is $-\infty$), then the dual problem must be infeasible*
- *If the dual problem is unbounded (i.e., the optimal value is ∞), then the primal problem must be infeasible*

Optimality conditions for LP: If $\mathbf{x}$, $\mathbf{y}$ satisfy:

1. $\mathbf{x}$ is primal feasible
2. $\mathbf{y}$ is dual feasible
3. The objective values are the same, i.e., $\mathbf{c}^T\mathbf{x} = \mathbf{b}^T\mathbf{y}$

Then $\mathbf{x}$ and $\mathbf{y}$ are optimal solutions to the primal and dual problems respectively.

Theorem (Strong Duality Theorem)

If a linear program has a finite optimal solution, so does its dual, and the optimal values of the primal and dual are equal

If the primal problem has an optimal solution $\mathbf{x}^*$, then it must be associated with some optimal basis B such that $\mathbf{x}_B = A_B^{-1}\mathbf{b}$ ($\mathbf{x}_B$ is the basic part of $\mathbf{x}^*$). Also, when the simplex method terminates, the reduced costs

$$\mathbf{c}^T - \mathbf{c}_B^T A_B^{-1} A \geq 0 \qquad (1)$$

Now we define $\mathbf{y}^T = \mathbf{c}_B^T A_B^{-1}$. By (2), $A^T\mathbf{y} \leq \mathbf{c}$, i.e., $\mathbf{y}$ is feasible to the dual problem. In addition

$$\mathbf{b}^T\mathbf{y} = \mathbf{c}_B^T A_B^{-1}\mathbf{b} = \mathbf{c}_B^T\mathbf{x}_B = \mathbf{c}^T\mathbf{x}^*$$

	Finite Optimum	Unbounded	Infeasible
Finite Optimum	√		
Unbounded			√
Infeasible		√	√

Theorem

Let $\mathbf{x}$ and $\mathbf{y}$ be feasible solutions to the primal and dual problems respectively. Then $\mathbf{x}$ and $\mathbf{y}$ are optimal solutions if and only if

$$\begin{aligned} x_i > 0 \quad &\Longrightarrow \quad A_i^T\mathbf{y} = c_i \\ A_i^T\mathbf{y} < c_i \quad &\Longrightarrow \quad x_i = 0\end{aligned}$$

Or in other words,

$$x_i \cdot (c_i - A_i^T\mathbf{y}) = 0, \quad \forall i.$$

Sometimes, we write the dual problem (equivalently) as

Primal		Dual	
min	$\mathbf{c}^T\mathbf{x}$	max	$\mathbf{b}^T\mathbf{y}$
s.t.	$A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0$	s.t.	$A^T\mathbf{y} + \mathbf{s} = \mathbf{c}, \quad \mathbf{s} \geq 0$

We call $\mathbf{s}$ the *dual slack* variables. Then the complementarity conditions can be written as

$$x_i \cdot s_i = 0 \quad \forall i$$

We sometimes call the complementarity conditions the *complementary slackness condition*.

Theorem

Let $\mathbf{x}$ and $\mathbf{y}$ are feasible solutions to the primal and dual problems respectively. Then $\mathbf{x}$ and $\mathbf{y}$ are optimal if and only if

$$y_i \cdot (\mathbf{a}_i^T\mathbf{x} - b_i) = 0, \quad \forall i; \qquad x_j \cdot (A_j^T\mathbf{y} - c_j) = 0, \quad \forall j.$$

Suppose after some iterations, it reaches an optimal solution with basis B. Then the tableau becomes:

$\mathbf{c} - \mathbf{c}_B^T A_B^{-1} A$	$-\mathbf{c}_B^T A_B^{-1}$	$-\mathbf{c}_B^T A_B^{-1}\mathbf{b}$
$A_B^{-1}A$	A_B^{-1}	$A_B^{-1}\mathbf{b}$

Therefore, the opposite to the reduced cost corresponding to the original identity matrix part is the optimal dual solution.

Final tableau:

B	0	0	0	1/2	1	250
1	1	0	0	-1/2	1	50
3	0	0	1	1/2	-1	50
2	0	1	0	1/2	0	100

What is the dual optimal solution?

- $(p_1, p_2, p_3) = (0, -1/2, -1)$, with objective value -250

Caveat: If the problem is not derived from adding slack variables (therefore the initial top row is not the actual $\mathbf{c}$), then this method may not give the right answer.

- In that case, one can just compute $(A_B^{-1})^T\mathbf{c}_B$.

Proof. Consider the pair of primal-dual linear program:

Primal		Dual	
min	$\mathbf{c}^T\mathbf{x}$	max	0
s.t.	$A\mathbf{x} = 0, \mathbf{x} \geq 0$	s.t.	$A^T\mathbf{y} \leq \mathbf{c}$

If there exists $\mathbf{x}$ such that

$$A\mathbf{x} = 0, \quad \mathbf{x} \geq 0, \quad \mathbf{c}^T\mathbf{x} < 0$$

Then by scaling this $\mathbf{x}$, we can make $\mathbf{c}^T\mathbf{x}$ arbitrarily negative, therefore, the primal problem must be unbounded.

However, for an LP, if the primal problem is unbounded, then the dual problem must be infeasible. Thus such an $\mathbf{x}$ is a *certificate* that the linear system $A^T\mathbf{y} \leq \mathbf{c}$ is infeasible. □

Using this transformation, we can write down the LP formulation. Let x_{ij} denote the amount of flow across edge (i,j).

max flow problem

$$\begin{aligned}\text{maximize}_{\mathbf{x},\Delta} \quad & \Delta \\ \text{subject to} \quad & \textstyle\sum_{j:(j,i)\in E} x_{ji} - \sum_{j:(i,j)\in E} x_{ij} = 0, && \forall i \neq s,t \\ & \textstyle\sum_{j:(j,s)\in E} x_{js} - \sum_{j:(s,j)\in E} x_{sj} + \Delta = 0 && \text{node s} \\ & \textstyle\sum_{j:(j,t)\in E} x_{jt} - \sum_{j:(t,j)\in E} x_{tj} - \Delta = 0 && \text{node t} \\ & x_{ij} \leq w_{ij}, && \forall (i,j) \in E \\ & x_{ij} \geq 0, && \forall (i,j) \in E\end{aligned}$$

construct the dual problem:

$$\begin{aligned}\text{minimize} \quad & \sum_{(i,j)\in E} w_{ij} z_{ij} \\ \text{subject to} \quad & z_{ij} \geq y_i - y_j, && \forall (i,j) \in E \\ & y_s - y_t = 1 \\ & z_{ij} \geq 0\end{aligned}$$

The dual problem is equivalent to finding a subset S of vertices containing s but not t, that minimizes the weight of the cut, i.e.

$$\sum_{i\in S, j\notin S} w_{ij}$$

Theorem

If the dual has a unique optimal solution $\mathbf{y}^*$, then $\nabla V(\mathbf{b}) = \mathbf{y}^*$.

- If the dual optimal solution is not unique (or is unbounded or infeasible), then the gradient does not exist.
- If one changes b_i by a small amount Δb_i, then the change of the objective value will be $\Delta b_i y_i^*$

Theorem

If the primal problem has a unique optimal solution $\mathbf{x}^*$, then

If one changes c_i by a small amount Δc_i, then the change of the objective value will be $\Delta c_i x_i^*$

$$\begin{aligned}\text{maximize}_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq 0\end{aligned}$$

If at optimal $\mathbf{x}^*$, $\mathbf{a}_i^T\mathbf{x}^* < b_i$, then what happens if we change b_i?

- By the complementarity conditions, the corresponding dual variable y_i^* must be 0.
- Therefore, changing the right-hand-side of an inactive constraint by a small amount won't affect the optimal value (also the optimal solution)
- Intuition: If a resource is already redundant, then adding or reducing a small amount wouldn't matter

Suppose $\mathbf{b}$ becomes $\mathbf{b} + \Delta\mathbf{b}$. Now the basic solution corresponding to the original optimal basis is

$$\tilde{\mathbf{x}}_B = A_B^{-1}(\mathbf{b} + \Delta\mathbf{b}) = \mathbf{x}^* + A_B^{-1}\Delta\mathbf{b}$$

Note that the reduced cost $\mathbf{c}^T - \mathbf{c}_B^T A_B^{-1} A$ doesn't depend on $\mathbf{b}$.

- If $\tilde{\mathbf{x}}_B \geq 0$, then B is still the optimal basis, and the new optimal solution is $(\tilde{\mathbf{x}}_B, 0)$ with the new optimal value

$$V(\tilde{\mathbf{b}}) = V^* + \mathbf{c}_B^T A_B^{-1}\Delta\mathbf{b} = V^* + (\mathbf{y}^*)^T\Delta\mathbf{b}$$

where $\mathbf{y}^*$ is the optimal dual solution (this explains the local theorem).

- If the original basis is still optimal, then the local sensitivity analysis holds.

Assume $\Delta\mathbf{b} = \lambda\mathbf{e}_i$ ($\mathbf{e}_i$ is a vector with 1 at position i). Then we need to have

$$\mathbf{x}^* + \lambda A_B^{-1}\mathbf{e}_i \geq 0$$

The final simplex tableau is

B	0	0	0	1/2	1	250
1	1	0	0	-1/2	1	50
3	0	0	1	1/2	-1	50
2	0	1	0	1/2	0	100

Thus $A_B^{-1} = \begin{pmatrix} 0 & -0.5 & 1 \\ 0 & 0.5 & 0 \\ 1 & 0.5 & -1 \end{pmatrix}$. If $\mathbf{b}$ changes to $\mathbf{b} + \lambda \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ then

$$\tilde{\mathbf{x}}_B = \mathbf{x}_B^* + \lambda A_B^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 50 \\ 100 \\ 50 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Case 1: $j \in B$

In this case, the reduced costs are

$$\begin{aligned} & \mathbf{c}_N^T - (\mathbf{c}_B^T + \lambda\mathbf{e}_j^T)A_B^{-1}A_N \\ = \ & \mathbf{c}_N^T - \mathbf{c}_B^T A_B^{-1}A_N - \lambda\mathbf{e}_j^T A_B^{-1}A_N\end{aligned}$$

Note that $\mathbf{c}_N^T - \mathbf{c}_B^T A_B^{-1}A_N$ is the reduced costs for the original problem. We denote it by $\mathbf{r}_N^T$. Therefore, in order to maintain the optimality of the current basis, we need to have

$$\mathbf{r}_N^T - \lambda\mathbf{e}_j^T A_B^{-1}A_N \geq 0 \qquad (1)$$

Case 2: $j \in N$

In this case, the reduced costs are:

$$\mathbf{c}_N^T + \lambda\mathbf{e}_j^T - \mathbf{c}_B^T A_B^{-1}A_N = \mathbf{r}_N^T + \lambda\mathbf{e}_j^T$$

Therefore, in order to maintain the optimality of the current basis, we need to have

$$\mathbf{r}_N + \lambda\mathbf{e}_j \geq 0 \qquad (2)$$

$A_B^{-1} = \begin{pmatrix} 0 & -0.5 & 1 \\ 1 & 0.5 & -1 \\ 0 & 0.5 & 0 \end{pmatrix}$; $A_N = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$; $\mathbf{r}_N = \begin{pmatrix} 0.5 \\ 1 \end{pmatrix}$

Assume we change the profit 1 from 1 to $1 + \lambda$ (i.e., $-1 - \lambda$ in the standard form). Then we need

$$\mathbf{r}_N - \lambda A_N^T (A_B^{-1})^T \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 1 \end{pmatrix} - \lambda \begin{pmatrix} 0.5 \\ -1 \end{pmatrix} \geq 0$$

Therefore, $-1 \leq \lambda \leq 1$

If the change of $\mathbf{c}$ is so much that the reduced cost of the current solution contains negative number, then

- We can continue with the simplex tableau until it reaches optimal solution.

If the change of $\mathbf{b}$ is so much that the solution corresponding to the original optimal basis B is no longer feasible

- Then we may need to solve the problem from the start
- However, it can be viewed as that the objective coefficients of the dual problem changed. Then one can use the method that deals with changes in objective coefficients

If the change is for a number in a non-basic column, say A_j, then the original optimal solution is still feasible, the only change is to the reduced cost of jth variable.

- Recompute $\bar{c}_j$. If it is still nonnegative, then the original optimal solution stays optimal. Otherwise, update the tableau for the jth column as well as the reduced cost and continue from there.

If the change is for a number in a basic column, then nearly all the numbers in the tableau will change. In general, there is not a simple way to deal with it.

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